Split Combine oLPN Process

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This document describes the process of splitting and combining oLPN to reduce shipping costs. Combining the oLPN process lets the user to combine two oLPNs of the same order. The splitting of the oLPN process lets users to split one oLPN into multiple oLPNs. This provides an ability for facilities to reduce the shipping cost when the oLPN is overloaded and shipping cost is higher based on the weight or so.

Configuration Summary

Split Combine oLPN

- Split Combine oLPN Strategy
- Split Combine oLPN Criteria

Process Flow

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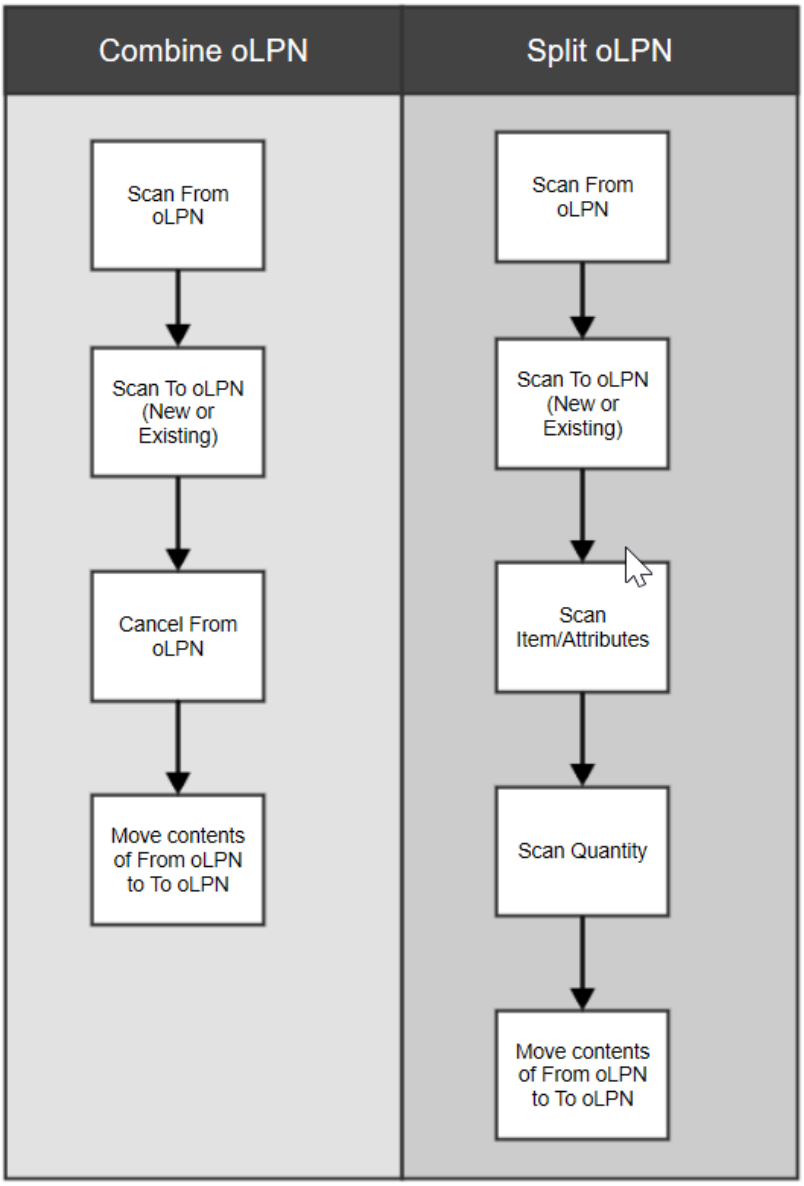

Retrieve Configuration

When the user selects a transaction, the system will retrieve the split combined oLPN strategy. Based on the rules defined, the system will identify the split combined oLPN criteria to determine the parameters required to process the oLPN scanned.

Accept From oLPN

This screen is the landing screen where the system will let users scan the "From oLPN" to combine or split. This screen will let users scan oLPN or oLPN cross reference.

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Validations

- If the "From oLPN" scan is invalid, the system checks if there is an oLPN cross reference oLPN scan. If both are invalid, then the error is "Invalid oLPN scanned".
- If the "From oLPN" scan is not in packed or created status, then the error goes out as "Invalid oLPN status".
- If the "From oLPN" scan is in created status and picking has started, then the split is not allowed. An error message is shown saying "Picking in progress"
- If the "From oLPN" scan belongs to an MPS group then system shows a warning message to user:
  - Warning message: oLPN is already planned in an MPS group. Splitting this oLPN would cancel the entire MPS group planning. Do you want to continue?
  - The user can accept the warning and continue the execution.
  - The default level of this message is warning.
  - If the oLPN is cancelled during the split process, then the entire MPS group will be cancelled or voided.

Updates

- The system generates a labor start message.

Next Steps

- Go to Accept To oLPN screen.

Accept To oLPN

This screen lets users to scan the "To oLPN" to combine or split. This screen lets users to scan "To oLPN" or oLPN cross reference. In case of combined oLPN, users have to scan the existing "To oLPN" and split transactions, letting users move inventory to the existing or new oLPN.
If the system generating oLPN is set ON, then "To oLPN" screen is skipped.

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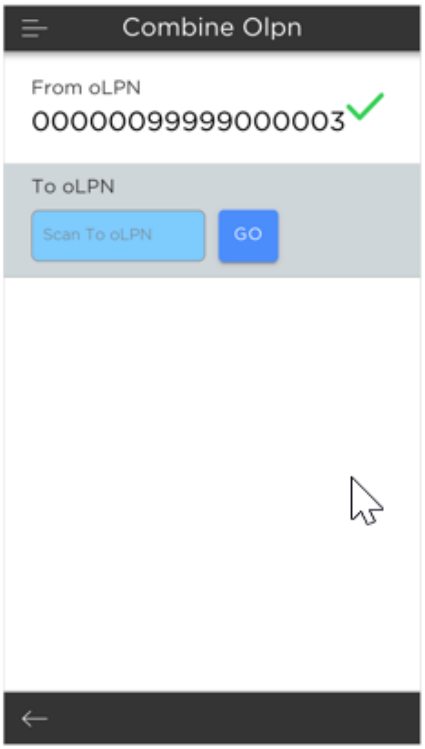

Validations

- If "To oLPN: scanned is invalid, the system checks for oLPN cross reference. If both are invalid, then the system performs Barcode type validation. If an invalid oLPN is barcode scanned, then error out "Barcode scanned is invalid. No barcode type defined". If oLPN is invalid, then the error goes out as "Invalid oLPN scanned".
- When the oLPN scanned in the *To oLPN* field is not in *Packed* status, the following error is displayed:
  "From oLPN status and To oLPN status do not match."
  However, if the oLPN status is configured as Shipped or Canceled in the Pick Pack parameter, "*What is the minimum status of the oLPNs that can be reused?*", the solution allows users to combine the oLPN with the shipped or canceled oLPN. If "*What is the minimum status of the oLPNs that can be reused?*" is not configured, then the solution allows users to combine the oLPN with a canceled oLPN but not with a shipped oLPN.
- If both From oLPN and To oLPN's order are different, then the error is "From oLPN order and To oLPN order do not match."
- If both From oLPN and To oLPN's shipment are different, then the error is "From oLPN shipment and To oLPN shipment do not match."
- If both From oLPN and To oLPN's stop are different, then the error is "From oLPN stop and To oLPN stop do not match."
- If both From oLPN and To oLPN's ship via are different, then the error is "From oLPN ShipVia and To oLPN ShipVia do not match."
- If the From oLPN is of Created status, then the system allows to split on to a new oLPN only.
- If break attribute is configured and the from oLPN and to oLPN have different values for the configured break attribute, an error is displayed saying "Can't combine oLPNs as break attribute has different values". E.g. if order planning run ID is configured as break attribute, then combining is allowed only if both from and to oLPNs have the same order planning run ID.
- If the "To oLPN" scan belongs to an MPS group, then the system can show a warning message to the user:
  - Warning message: oLPN is already planned in an MPS group. Combining this oLPN would disrupt the entire MPS group planning. Do you want to continue?

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- The user can accept the warning and continue the execution.
- The default level of this message is Non-display.

Prevent SplitCombine of oLPN's with condition codes.

This is validated when the From oLPN and To oLPN are scanned during the splitcombine transaction.

- If the From oLPN or To oLPN has condition codes applied and the splitcombine criteria parameter "Do you want to prevent split/combine of containers having Condition codes applied? " is set to Y , splitcombine of oLPN's having condition codes can be restricted.
- If we want to allow the splitcombine of containers with certain condition codes, then these condition codes should be configured and set in the override condition code group of splitcombine criteria. [What is Override Condition Code Group for splitcombine? -should be configured with a valid condition code group.]

Updates

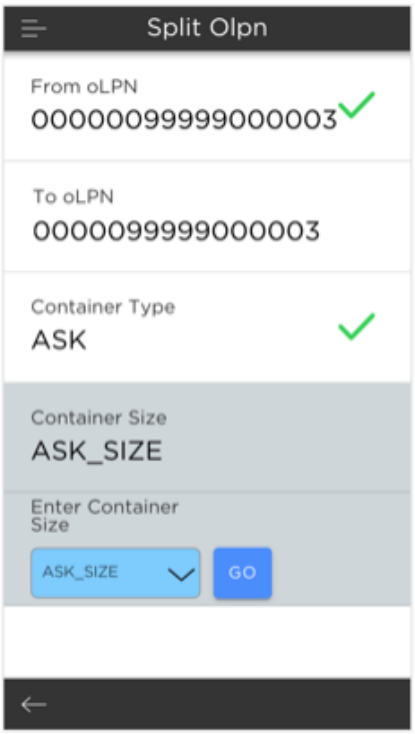
- No Updates

Next Steps

- If the accepted container type size is set ON, then the system goes to "Accept Container Type Size".

Accept Container Type Size

This screen lets users to scan the container type size for the new To oLPN. If the To oLPN is an existing oLPN, then system skips this screen. Container type size is accepted only for new To oLPN. Default container type and size are pre-populated if configured. If the default container type size is configured to the criteria and accepted container type size is set OFF, the new To oLPN is created with the default container type size configured in the criteria.
If the accepted container type size is set OFF, then the container type size screen is skipped.



Validations

- If the container type is invalid, then the error is "Invalid container type".
- If the container size is invalid, then the error is "Invalid container size".

Updates

- No Updates

Next Steps

- If the accepted oLPN cross reference is set ON, then the system goes to "Accept oLPN Cross Reference".

Accept oLPN Cross Reference

This screen lets users to scan the oLPN cross reference for the new To oLPN. If the To oLPN is an existing oLPN, to the system skips this screen. oLPN cross reference is accepted only for new To oLPN.
If the accepted oLPN cross reference is set OFF, then oLPN cross reference screen is skipped.

Validations

- If the scanned oLPN cross reference does not match the barcode definition, then the error is "Barcode scanned is invalid. oLPN barcode type defined".
- If the scanned oLPN cross reference already exists, then the error is "Scanned oLPN cross reference already exists".
- If an oLPN ID exists with the same oLPN cross reference scanned, then the error is "Invalid oLPN cross reference".

Updates

- If Verify Item is set to "None", then perform the updates mentioned in the **Updates** session. Go back to the "Accept From oLPN" screen.
- The system also generates labor detail messages for each item transferred to To oLPN.

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Next Steps

- If Verify Item is set to "Item" OR "Item Pack Single Detail", then go to "Accept Item".

Accept Item

This screen lets users to scan the item in case of split oLPN transaction. If a verified item is configured as an "Item" or "Item Pack Single Detail", then the system accepts the item from the user.

Validations

- If the entered item barcode is invalid, then the error is "Barcode scanned is invalid. Item barcode type is defined".
- If the scanned item is an invalid item, then the error is "Invalid Item".
- If the scanned item is not part of From oLPN, then the error is "Item scanned is not part of From oLPN".

Updates

- If the Verify Item is set to "Item" and the verified quantity is set OFF, then perform updates for the 1 unit that is moved from From oLPN to the To oLPN. See **Updates** session for more details.
- If the Verify Item is set to "Item Pack Single Detail", then perform updates to move all the inventory of the item scanned from the From oLPN to the To oLPN. See **Updates** session for more details.
- The system also generates labor detail messages for each item transferred to To oLPN.

Next Steps

- If Verify Item is set to "Item" and verified quantity is set ON, then go to "Accept Quantity" screen(when the item does not track item attributes or attributes are not being captured.)
- If Verify Item is set to "Item" and capture "Item Attributes" is set ON, then go "Accept Item Attributes" screen.

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Accept Item Attributes

This screen lets users scan the item attributes tracked at the item level. The solution prompts users to scan attributes based on the "*Do you want to capture the Item Attributes?"* parameter configured in the Split Combine Criteria screen.
Additionally, the system allows the prevention of capturing certain item attributes, provided those attributes have the same values across all the details of the oLPN for a given item. This can be achieved by configuring those attributes in the Item Attribute Override template defined in the Split Combine criteria.

> **Note**: The prompt behavior for Item attributes supports only Always Prompt or Prompt is NULL during the split/combine oLPN process.

Users cannot edit attributes that do not belong to the oLPN detail. Any attribute mismatch displays the error message: PPK::0089 Item Attribute Mismatch.
When combining two oLPNs, the item attribute prompt is not required, and attribute values are merged into the oLPN. Configure the *Do you want to capture the Item attributes* parameter to accept attributes during Split oLPN.
The above configuration lists the options for selection:

- Do not capture(default value)
- Accept from user(The attribute fields are defaulted empty and user is prompted to scan attributes)
- Accept from user, default from allocated-(Display the attribute values defaulted from oLPN detail, user can choose to accept or edit the attribute)

Next Steps

- If Verify Quantity is set ON, then go to the "Accept Quantity" screen.

Accept Quantity

This screen allows users to enter the quantity that needs to be transferred to the **To oLPN**.

- The quantity entered follows the **Quantity UOM** defined in the criteria. If no Quantity UOM is configured, the system will default to the item's **base UOM (unit).**
- If the split item contains Unique Attributes (such as serial numbers), the system determines the Display UOM, Remainder UOM, and their respective quantities based on the UOMs used when those attributes were originally packed on the oLPN being split.

Example:
An oLPN contains a serial-tracked item with serials captured in both **Packs** and **Units**.
The quantity mapping is as follows:

- **1 Pack = 5 Units**
- **Total oLPN Quantity** = 23 Units
- **Serials Captured**: 4 Packs and 3 Units

In this scenario, the system will display:

- **Display UOM**: Pack and **Quantity** as 4
- **Remainder UOM**: Unit and **Quantity** as 3

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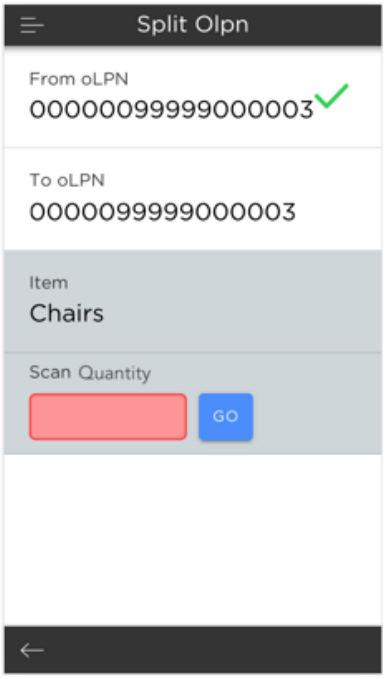

Validations

- If the quantity is not entered, then the error is "Quantity must be greater than zero".
- If the quantity entered is greater than oLPN quantity for the item, then the error is "Transfer quantity cannot be greater than From oLPN quantity".

Updates

- Perform transfer of inventory for the quantity scanned. See **Updates** session for more details.
- The system also generates labor detail messages for each item transferred to To oLPN.

Next Steps

- Go back to Accept From oLPN screen.

Accept Attributes

The system displays the attribute capture screen based on whether the item tracks unique or non-unique attributes. For instance, if an item is serial-tracked, the screen will prompt the user to capture the required serial numbers.

Serial Number capture

- When part of an oLPN is moved from a source to a destination, the prompt for capturing unique attributes is driven by the **Units of Measure (UOMs)** in which those attributes were originally packed on the oLPN.
  Example:
  An oLPN contains a serial-tracked item with serials recorded in both **Packs** and **Units**.
  - 1 Pack = 5 Units
  - Serials captured: 4 Packs and 3 Units
  - Total oLPN quantity = (4 Packs × 5 Units) + 3 Units = 23 Units
  - In this scenario, the system allows splitting inventory in terms of both packs and units. Serial numbers will be captured for the split quantities according to their corresponding UOMs.

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- In **Unit Scan** mode, the serial number prompt appears after each barcode scan.
- If the entire oLPN is moved to a new or existing oLPN, the system does not prompt for serial capture since all serials are transferred together.

Validations

- The serial number entered should belong to the oLPN being split.

Updates

- The serial numbers are updated at oLPN level in the pick pack component.
- The serial number movement is also captured in inventory.

Catch Weight Capture

- Catch weight capture is controlled by an attribute called "Non Unique Attribute Capture Mode Id". Catch weight can be either calculated or captured using this attribute.
- Catch weight can be captured at various UOMs like Pack, LPN, unit etc.
- When a part of oLPN is moved from a source to destination, the catch weight is captured/calculated for both source and destination oLPN
- When the complete oLPN is split into a new oLPN or combined with an existing oLPN, then the catch weight prompt is not displayed as all the catch weights are moved from source to destination.
- Catch weight entry is not supported in unit scan mode.

Validations

- If the entered catch weight is beyond the defined tolerance for the item, then an error message is displayed. Validations can be configured for both upper and lower tolerance in the attribute group.

Updates

- Catch weight entries are created for the newly created oLPNs if a new oLPN is created.
- For the existing oLPNs, the catch weight entry is updated as per the user entry or calculation.
- If the oLPN gets cancelled as a part of combined oLPN or split oLPN, the corresponding catch weight entries are removed.
- Catch weight is also updated on the inventory side.

Task detail updates

- If the oLPN is in created status, then the pick/pack task detail or pack task detail is also split and a new task detail is created based on the entered quantity.

Accept Printer Location

This screen appears when 'PrinterPromptId' is set as 'Prompt for Printer'. Depending on the 'FromOlpnPrintCriteria' and 'ToOlpnPrintCriteria', a request is sent for the respective oLPNs to print components so that the labels are generated.

Validations

- Once the location is scanned, validate if the location has a 'Printer Zone' configured. If it is present, accept the location, else an error is displayed.

Updates

- LM messages are written for this user with this transaction to provide credit for travelling to the location where the printer is set up.
- A request is sent to the printing component with the appropriate container ID for label generation.

End oLPN

End oLPN action is available on the Item and quantity screen. This will let users to end the split oLPN process and perform the updates. On End oLPN, the system performs the updates and takes the user to From oLPN screen. The system also sends a labor end message at this point.

Updates

From oLPN Updates

oLPN Detail

oLPN Detail Attributes	Comments
Initial Quantity	Reduce by scanned quantity
Picked Quantity	Reduce by scanned quantity if pickedQuantity > 0
Packed Quantity	Reduce by scanned quantity
Status	If initial quantity = 0, then delete the oLPN details.

oLPN Header

oLPN Attributes	Comments
Status	Re-calculate the oLPN status. If all the oLPN details are canceled, then update the status to Canceled/Delete.
Current Location	If oLPN is cancelled/ deleted, then clear it out. Update the current quantity of location also.
Destination Location	If oLPN is cancelled/ deleted, then clear it out. Cancel the corresponding tasks or update the location directed quantity.
Estimated Weight	Re-calculate the estimated weight of the oLPN as the quantity is reduced.
Estimated Volume	Volume is untouched as the volume comes from container type/ size. If container type size = Null and estimated volume = Null, then calculate based on the unit item volume.
Total LPN Quantity	Re-calculate the total LPN quantity.
Is Single Line	Re-calculate this flag. If there is only a single item in the oLPN, then update this flag as Yes.
Weight	Clear this out as this oLPN has to be re-weighed.
Volume	No change
Pallet	Pallet ID needs to be cleared. oLPN has to be removed from Pallet.
LPN Creation Code	No updates

Other From oLPN Updates

If all the oLPN details are canceled, then oLPN needs to be canceled. (See "**Cancel oLPN**" updates for more details)

If a partial quantity gets moved, then:

- If the Routing strategy is configured, then make a call to Routing with the From oLPN and Routing Strategy to perform routing updates.

- If the oLPN is parcel and ship via has changed, then call the parcel to get the new tracking number.
- Re-print the From oLPN labels if the "From oLPN Print Criteria" is configured.

To oLPN Updates

oLPN Detail

oLPN Detail Attributes	Comments
Initial Quantity	Increase in scanned quantity
Picked Quantity	Do not update it.
Packed Quantity	Increase in scanned quantity
Status	Packed status

oLPN header

oLPN Attributes	Comments
Status	Calculate the oLPN status.
oLPN Type / Size	In the case of a new oLPN, use whatever user has entered or the default container type/size configured on criteria. If not configured, inherit the type and size from the From olpn and stamp it on the To olpn.
Estimated Weight	Re-calculate the estimated weight for the To oLPN (similar to when a new oLPN is created). Add the item unit weight or order line unit weight for the new quantities added to this oLPN.
Estimated Volume	Volume is taken from container size/ size. If container type size = Null and estimated volume = Null, then calculate based on the unit item volume.
Total LPN Qty	Re-calculate the total LPNQty.
Is Single Line	If the oLPN has more than one Item, then update this Flag OFF. If not updated, keep it as OFF.
Weight	Clear out weight as oLPN Status will be updated to Packed status. To oLPN has to be weighed again.
Volume	Use the OLPN Type/Size volume.
Pallet	Pallet ID will be null, irrespective of the original oLPN is on a Pallet or not.
LPN Creation Code	7100 - If a new oLPN is created during the split process.
Shipment	Copy it from "From oLPN" if it is a new oLPN OR determined based on the Routing Strategy, if configured.
Shipvia	Copy it from "From oLPN" if it is a new oLPN OR determined based on the Routing Strategy, if configured.
Stop	Copy it from "From oLPN" if it is a new oLPN OR determined based on the Routing Strategy, if configured.
Static Route	Copy it from "From oLPN" if it is a new oLPN OR determined based on the Routing Strategy, if configured.

Other To oLPN Updates

- If Routing Strategy is configured, then call routing with To oLPN ID and Routing Strategy to perform routing updates.
- If the To oLPN is parcel, then call parcel to get the tracking number.
- Print the To oLPN labels, if the "To oLPN Print Criteria" is configured.

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- If the To oLPN status = Packed, generate MHE event for the To oLPN in case of the new oLPN (oLPN MHE Divert).
- If the criteria parameter "Copy the condition codes of FROM oLPN to TO oLPN during Split/Combine process" is configured as Yes, system will copy the condition codes of the From oLPN to To oLPN.

LM Interactions

Events during Split Combine oLPN

WM publishes LM events at various touch points while doing the Split Combine oLPN process.

Labor Event Message	Trigger point	LM Message Details
LM Start event	Event is triggered when user logs into the Mobile transaction and prompted with From oLPN scan to initiate the Split Combine oLPN process.	Contains information on: • Transaction • User who is performing the process • Start time of operation • LM Activity Name
LM Detail event	Event is triggered whenever user transfers inventory into the new To oLPN. If there are multiple items being transferred into the oLPN, then each adjustment will trigger one LM event with details on the Item, attributes, and quantity being adjusted. **Note:** Task and LM event details are updated based on the From oLPN. The quantity on each detail reflects the quantity transferred from the From oLPN to To oLPN. For example, if only one item is moved from the From oLPN to To oLPN and the From oLPN is canceled, then the labor activity quantity for the transferred qty is 1.	Contains the following information: • Transaction • To oLPN • Item and corresponding attributes being adjusted • Transferred quantity.
LM End event	Event is triggered when a combined oLPN is complete OR whole From oLPN split is complete OR when user performs End oLPN.	Contains information on: • Transaction Id • User who is performing the process • End time of operation • LM Activity Name
LM End event	The event is triggered even when the user exits the transaction by clicking on the Menu option.	

Activity Tracking

Split combined activity tracking is mainly logged under the following transaction types.

Transaction Type	Functional Flows
SPLIT_COMBINE_OLPN	Split oLPN Combine oLPN

The following attributes are captured during split, combined transactions for activity tracking.

Attributes	Description
Employee	User executing the Split/combine transaction
Container	Identifies the outbound LPN from which the split happened
New Container	Identifies the outbound LPN to which the split happened
Transaction type	SPLIT_COMBINE_OLPN

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Transaction	Name of the transaction from where split/combine was invoked

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